

Nikunj Vaghasiya

Computer Engineering Student | Cybersecurity & AI/ML Developer

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PROFESSIONAL SUMMARY

Driven Computer Engineering student at **CHARUSAT** (CGPA: 7.53) specializing in **Cybersecurity**, **AI-driven development**, and **full-stack web engineering**. Hands-on experience building scalable Django web applications, end-to-end encrypted communication systems, and real-time threat detection tools. Proven ability to bridge software development with security research. Actively solving penetration testing labs on **TryHackMe** and **Hack The Box**. Seeking to contribute to innovative, security-first engineering teams.

TECHNICAL SKILLS

Languages	C, C++, Java, Python, SQL, HTML, CSS
Web & Frameworks	Django, Django REST Framework, Flask, React, Socket.IO, REST APIs
Cybersecurity	Metasploit, Burp Suite, Wireshark, Nmap, Honeypots, OWASP Top 10, AES/RSA Encryption, SSH Hardening, Privilege Escalation, PortSwigger Labs
AI & Data Science	OpenCV, Deep Learning (CNNs), NLP, Voice Recognition, Scikit-Learn, TensorFlow/Keras, Pandas, NumPy
Tools & DevOps	Git, GitHub, Linux/Bash, Docker, Postman, MySQL, SQLite, Paramiko
Platforms	TryHackMe, Hack The Box, HackerRank, LeetCode

PROJECTS

DocGuard AI	<i>Current Python · CNNs · NLP · Django REST API</i>
- Building an AI framework for document tampering detection using ML-based verification pipelines. - Leveraging CNNs and NLP models to flag forged documents in real-time with high accuracy. - Integrated a Django REST API backend to serve model predictions and manage document workflows.	
EduHub – Online Education Platform	<i>Django · React · MySQL</i>
- Designed and deployed a scalable Learning Management System with secure JWT authentication. - Implemented class creation, assignment management, and real-time student-teacher communication. - Built RESTful APIs with Django REST Framework consumed by a React frontend.	
Ransomware Detection Engine	<i>Python · Behavioral Analysis · OS Hooks</i>
- Developed a real-time behavioral analysis tool detecting encryption patterns and unauthorized file-system changes. - Monitored I/O events via OS-level hooks to flag ransomware activity before encryption completes.	
SSH Honeypot for Intrusion Detection	<i>Python · Paramiko · Network Security</i>
- Implemented a low-interaction SSH honeypot to capture and log real-time brute-force attack patterns. - Automated log-parsing scripts to visualize attacker behavior and generate threat-intelligence reports.	
Secure End-to-End Messaging System	<i>Python · Socket.IO · AES/RSA</i>
- Built a chat application with end-to-end encryption ensuring zero plaintext transmission over the wire. - Implemented hybrid AES + RSA key exchange for scalable, secure multi-user sessions.	
AI Desktop Assistant	<i>Python · NLP · Voice Recognition</i>

- Built an AI-powered voice assistant with NLP capabilities for task automation including web search and app control.
- Integrated speech-to-text and intent-classification pipelines for natural language command processing.

Advanced Face Detection System

Python · OpenCV · Deep Learning

- Created a real-time facial recognition system using OpenCV with Haar cascades and CNN-based classification.
- Optimized inference pipeline for low-latency detection on standard hardware.

Deaf & Blind Accessibility Bridge

Python · OpenCV · NLP · Voice Recognition

- Engineered an AI/ML accessibility tool enabling gesture-to-text and text-to-speech translation for differently-abled users.
- Integrated real-time hand-gesture classification using a trained CNN model with 90%+ accuracy.

EDUCATION

B.Tech in Computer Engineering

Charotar University of Science & Technology (CHARUSAT), Gujarat

2023 – Present

CGPA: **7.53**

Relevant Coursework: Cloud Computing, Networking, Cybersecurity, DBMS, Operating Systems, DSA

Higher Secondary Education (HSC)

Fareni Gurukul, Dhoraji

2021 – 2023

Percentile Rank: **92.86 PR**

CERTIFICATIONS

- **Google** – Cybersecurity Professional Certificate
 - **HackerRank** – Java, Python, SQL, Problem Solving
- **Cisco** – Networking Fundamentals Certificate
 - **Infosys Springboard** – Java Programming Fundamentals